

TOPIC 1: INTRODUCTION TO INTERMEDIATE SCIENCE

1. Meaning of Science

Science is the systematic study of the natural world through observation, experimentation and evidence. It is a way of acquiring knowledge about living and non-living things and the processes that occur around us.

Intermediate Science at Junior Secondary level introduces students to basic scientific concepts, methods and the application of science to everyday life and the environment.

2. Importance of Science

- It helps us understand the world around us.
- It improves the quality of life through inventions (medicine, transport, communication).
- It solves problems such as disease, food shortage and environmental challenges.
- It promotes critical thinking and logical reasoning.
- It drives technological and economic development.
- It helps us protect and conserve the environment.

3. Branches and Scope of Science

a) Biology (Life Science) – study of living things (plants, animals, humans, microorganisms).

b) Chemistry – study of matter, its properties, composition and the changes it undergoes.

c) Physics – study of energy, forces, motion, light, sound, electricity and magnetism.

d) Earth / Environmental Science – study of the earth, weather, climate, soil, water and the environment.

Intermediate Science at JSS level combines elements of these branches with a strong focus on the environment and everyday applications.

4. The Role of Science in Everyday Life

- Cooking and food preservation involve chemical changes and hygiene.
- Medicine and health care depend on scientific research.
- Transportation (cars, planes) and communication (phones, internet) are products of science.
- Farming uses scientific knowledge of soil, weather and improved seeds.
- Electricity, clean water and waste management rely on scientific principles.
- Understanding weather helps us plan daily activities and farming.

5. Attitudes of a Good Scientist

A good scientist should show the following attitudes:

- **Curiosity** – desire to ask questions and find out how things work.
- **Honesty** – reporting observations and results truthfully.
- **Open-mindedness** – willingness to accept new evidence and change ideas.
- **Objectivity** – basing conclusions on facts, not personal feelings.
- **Patience and Perseverance** – continuing experiments even when results are slow.
- **Critical thinking** – analysing information carefully before drawing conclusions.
- **Respect for evidence** – accepting what experiments and observations show.
- **Safety consciousness** – following laboratory and environmental safety rules.

6. Summary

Science is the systematic study of nature. It is important for understanding the world, solving problems and improving life. Its main branches are Biology, Chemistry, Physics and Environmental Science. A good scientist

is curious, honest, objective, patient and safety-conscious.

End of Topic 1 – Introduction to Intermediate Science